

Shortest path in Directed Acyclic Graph

Solved

Difficulty: **Medium** Accuracy: **48.48%** Submissions: **236K+** Points: **4** Average Time: **20m**

Given a Directed Acyclic Graph of V vertices from 0 to $V-1$ and a 2D Integer array(or vector) `edges[ ][ ]` of length E , where there is a directed edge from `edge[i][0]` to `edge[i][1]` with a distance of `edge[i][2]` for all i .

Find the **shortest** path from **src(0)** vertex to all the vertices and if it is impossible to reach any vertex, then return **-1** for that vertex.

Examples :

Input: $V = 4$, $E = 2$, `edges = [[0,1,2], [0,2,1]]`

Output: `[0, 2, 1, -1]`

Explanation: Shortest path from 0 to 1 is 0->1 with edge weight 2. Shortest path from 0 to 2 is 0->2 with edge weight 1. There is no way we can reach 3, so it's -1 for 3.

Input: $V = 6$, $E = 7$, `edges = [[0,1,2], [0,4,1], [4,5,4], [4,2,2], [1,2,3], [2,3,6], [5,3,1]]`

Output: `[0, 2, 3, 6, 1, 5]`

Explanation: Shortest path from 0 to 1 is 0->1 with edge weight 2. Shortest path from 0 to 2 is 0->4->2 with edge weight 1+2=3. Shortest path from 0 to 3 is 0->4->5->3 with edge weight 1+4+1=6. Shortest path from 0 to 4 is 0->4 with edge weight 1. Shortest path from 0 to 5 is 0->4->5 with edge weight 1+4=5.